B3: Atomic and Laser Physics — MT 2020 Model Solutions

Question 1 — Central field, Thomas–Fermi atom

Problem. For an N-electron atom: (a) state the central-field approximation and the good quantum numbers; (b) sketch the effective potential of sodium; (c) using a Fermi-gas description ($n = 8\pi p^3/3h^3$) derive the Thomas–Fermi equation $\chi'' = \chi^{3/2}/x^{1/2}$ with $x = r/b$ and find b; (d) discuss its limits; (e) explain why $4s$ fills before $3d$ in K.

(a)

The two-body term $\sum_{i>j} e^2/(4\pi\varepsilon_0 r_{ij})$ prevents separation. Replace by spherical mean acting on each electron:

$$\sum_{i>j} \frac{e^2}{4\pi\varepsilon_0 r_{ij}} \longrightarrow \sum_i S(r_i).$$

Each electron sees a central potential:

$$U(r) = -\frac{Ze^2}{4\pi\varepsilon_0 r} + S(r).$$

Hamiltonian becomes a sum of one-electron terms; orbitals are labelled by single-particle quantum numbers:

$$\boxed{n,\ \ell,\ m_\ell,\ m_s}$$

Energies depend on n and ℓ (non-Coulombic). Residual electron repulsion couples individual ℓ_i, s_i so only L, S, J, M_J are strictly good for the full atom.

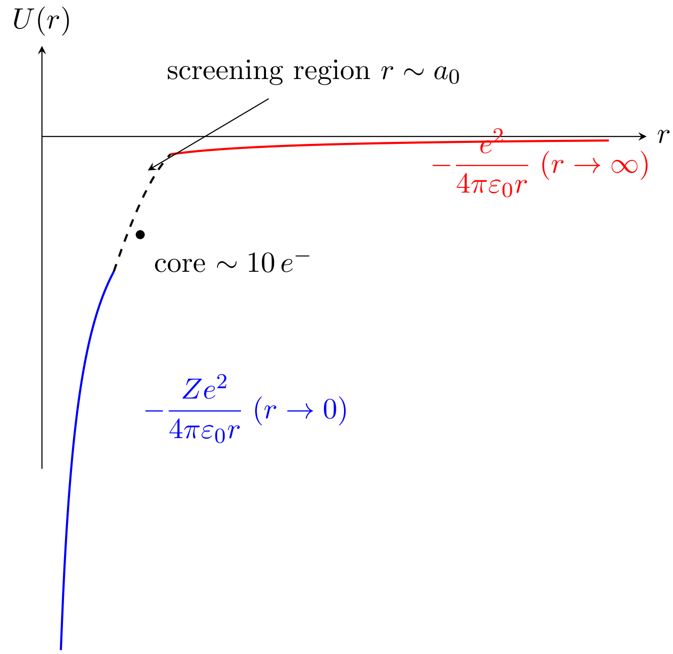
(b)

Two limits set the shape of $U(r)$ for Na ($Z = 11$, 10 core electrons):

$$r \to 0: \quad U(r) \to -\frac{Ze^2}{4\pi\varepsilon_0 r}.$$

$$r \to \infty: \quad U(r) \to -\frac{e^2}{4\pi\varepsilon_0 r}.$$

Screening turns on over $r \sim a_0$.



(c)

Let $V(r)$ be the electric potential of the screened nucleus, positive near origin. Electron PE is $-eV$. Just-bound condition $p_F^2/(2m) - eV = 0$:

$$p_F^2/(2m) = eV(r).$$

Solve for p_F:

$$p_F(r) = \left[2meV(r)\right]^{1/2}.$$

Insert into the given density $n = 8\pi p^3/(3h^3)$ with $p = p_F$:

$$n_e(r) = \frac{8\pi}{3h^3}\left[2meV(r)\right]^{3/2}.$$

Parametrise with screening function χ, $Z_{\text{eff}} = \chi Z$, BCs $\chi(0) = 1$, $\chi(\infty) = 0$:

$$V(r) = \frac{Ze\,\chi(r)}{4\pi\varepsilon_0 r}.$$

Multiply by e:

$$eV(r) = \frac{Ze^2\chi(r)}{4\pi\varepsilon_0 r}.$$

Substitute into n_e:

$$n_e(r) = \frac{8\pi}{3h^3}\left[\frac{2me^2 Z\chi}{4\pi\varepsilon_0 r}\right]^{3/2}.$$

Poisson's equation with electron charge density $\rho_e = -en_e$:

$$\nabla^2 V = -\rho_e/\varepsilon_0 = \frac{e\,n_e(r)}{\varepsilon_0}.$$

Use the radial identity:

$$\nabla^2 V = \frac{1}{r}\frac{\mathrm{d}^2(rV)}{\mathrm{d}r^2}.$$

Substitute $rV = Ze\chi/(4\pi\varepsilon_0)$:

$$\frac{Ze}{4\pi\varepsilon_0}\frac{1}{r}\frac{\mathrm{d}^2\chi}{\mathrm{d}r^2} = \frac{e}{\varepsilon_0}\frac{8\pi}{3h^3}\left[\frac{2me^2Z\chi}{4\pi\varepsilon_0 r}\right]^{3/2}.$$

Multiply both sides by $4\pi\varepsilon_0 r/(Ze)$:

$$\frac{\mathrm{d}^2\chi}{\mathrm{d}r^2} = \frac{4\pi r}{Z}\cdot\frac{8\pi}{3h^3}\cdot\frac{(2me^2Z)^{3/2}\chi^{3/2}}{(4\pi\varepsilon_0 r)^{3/2}}.$$

Cancel one Z against $Z^{3/2}$:

$$\frac{\mathrm{d}^2\chi}{\mathrm{d}r^2} = \frac{32\pi^2}{3h^3}\cdot\frac{(2m)^{3/2}e^3Z^{1/2}}{(4\pi\varepsilon_0)^{3/2}}\,\frac{\chi^{3/2}}{r^{1/2}}.$$

Collect powers of r and constants:

$$\frac{\mathrm{d}^2\chi}{\mathrm{d}r^2} = C\,\frac{\chi^{3/2}}{r^{1/2}}, \qquad C = \frac{32\pi^2}{3h^3}\,\frac{(2m)^{3/2}e^3Z^{1/2}}{(4\pi\varepsilon_0)^{3/2}}.$$

Set $r = bx$, so $\mathrm{d}^2/\mathrm{d}r^2 = (1/b^2)\mathrm{d}^2/\mathrm{d}x^2$:

$$\frac{1}{b^2}\frac{\mathrm{d}^2\chi}{\mathrm{d}x^2} = C\,\frac{\chi^{3/2}}{b^{1/2}x^{1/2}}.$$

Demand the universal form $\chi''(x) = \chi^{3/2}/x^{1/2}$:

$$\frac{1}{b^2} = \frac{C}{b^{1/2}}.$$

Multiply by $b^{1/2}$:

$$b^{-3/2} = C.$$

Invert:

$$b^{3/2} = \frac{1}{C}.$$

Take the 2/3 power:

$$b = C^{-2/3} = \left(\frac{3h^3(4\pi\varepsilon_0)^{3/2}}{32\pi^2(2m)^{3/2}e^3Z^{1/2}}\right)^{2/3}.$$

Apply the 2/3 to each factor:

$$b = \left(\frac{3h^3}{32\pi^2}\right)^{2/3}\frac{(4\pi\varepsilon_0)}{(2m)e^2Z^{1/3}}.$$

Pull out the Z dependence:

$$b = \left(\frac{3h^3}{32\pi^2}\right)^{2/3}\frac{4\pi\varepsilon_0}{2me^2}\,Z^{-1/3}.$$

Substitute $h = 2\pi\hbar$, so $h^3 = 8\pi^3\hbar^3$:

$$\frac{3h^3}{32\pi^2} = \frac{3\cdot 8\pi^3\hbar^3}{32\pi^2} = \frac{3\pi\hbar^3}{4}.$$

Raise to 2/3:

$$\left(\frac{3h^3}{32\pi^2}\right)^{2/3} = \left(\frac{3\pi\hbar^3}{4}\right)^{2/3} = \frac{(3\pi)^{2/3}}{4^{2/3}}\hbar^2.$$

Use the Bohr radius $a_0 = 4\pi\varepsilon_0\hbar^2/(me^2)$:

$$\frac{4\pi\varepsilon_0}{2me^2} = \frac{a_0}{2\hbar^2}.$$

Combine:

$$b = \frac{(3\pi)^{2/3}}{4^{2/3}}\hbar^2 \cdot \frac{a_0}{2\hbar^2}\, Z^{-1/3}.$$

Cancel $\hbar^2$:

$$b = \frac{(3\pi)^{2/3}}{2\cdot 4^{2/3}}\, a_0 Z^{-1/3}.$$

Reduce $2\cdot 4^{2/3} = 2^{1+4/3} = 2^{7/3} = (2^7)^{1/3} = 128^{1/3}$:

$$b = \frac{(3\pi)^{2/3}}{128^{1/3}}\, a_0 Z^{-1/3}.$$

Combine cube roots using $(3\pi)^{2/3} = (9\pi^2)^{1/3}$:

$$\boxed{b = \left(\frac{9\pi^2}{128\, Z}\right)^{1/3} a_0 \approx 0.885\, Z^{-1/3} a_0.}$$

(d)

Continuum Fermi-gas requires local Fermi wavelength $\ll$ scale on which V varies. Fails near the nucleus ($V \to +\infty$, density diverges), in the outer fringe (few electrons, statistics breaks), and for light atoms (low N). Reliable for the core of heavy atoms.

(e)

In a screened (non-Coulombic) central field, low-ℓ orbitals *penetrate* the core and feel a larger Z_{eff}. Centrifugal barrier $\ell(\ell+1)\hbar^2/(2mr^2)$ excludes $3d$ ($\ell = 2$) from the core, so $3d$ sees $Z_{\text{eff}} \approx 1$. The $4s$ orbital, although higher n, has substantial inner radial probability and sits at lower energy than $3d$. Therefore K is $[\text{Ar}]\,4s^1$.

Question 2 — Hyperfine structure in magnetic fields

Problem. Hamiltonian $\mathcal{H} = A\vec{I}\cdot\vec{J} + g_J\mu_B\vec{J}\cdot\vec{B} - g_I\mu_N\vec{I}\cdot\vec{B}$. (a) interpret each term; (b) estimate weak-field threshold; (c) derive $\Delta E = \frac{1}{2}AK + g_F\mu_B BM_F$; (d) draw level diagram for $J = 3/2, I = 3$; (e) strong-field hydrogen $1s\,{}^2S_{1/2}$, verify ratio.

(a)

$A\,\vec{I}\cdot\vec{J}$: magnetic hyperfine coupling between nuclear moment $\propto \vec{I}$ and the electronic field at the nucleus.

$g_J\mu_B\,\vec{J}\cdot\vec{B}$: electronic Zeeman coupling of $-g_J\mu_B\vec{J}/\hbar$ to $\vec{B}$.

$-g_I\mu_N\,\vec{I}\cdot\vec{B}$: nuclear Zeeman coupling; smaller than the electronic term by $\mu_N/\mu_B \sim m_e/m_p \approx 1/1836$.

(b)

Weak field requires hyperfine $\gg$ electronic Zeeman:

$$AI \gg g_J \mu_B B.$$

Solve for the crossover field:

$$B \ll B^* \equiv \frac{AI}{g_J \mu_B}.$$

Take alkali ground-state $A/h \sim 1\,\text{GHz}$:

$$A \sim h \cdot 10^9\,\text{Hz}.$$

Multiply $h = 6.626 \times 10^{-34}\,\text{J s}$:

$$A \approx 6.6 \times 10^{-25}\,\text{J}.$$

Plug $g_J \approx 2$, $I \sim 3/2$, $\mu_B = 9.27 \times 10^{-24}\,\text{J/T}$:

$$B^* \sim \frac{(6.6 \times 10^{-25})(1.5)}{(2)(9.27 \times 10^{-24})}\,\text{T}.$$

Reduce numerator:

$$(6.6 \times 10^{-25})(1.5) = 9.9 \times 10^{-25}\,\text{J}.$$

Reduce denominator:

$$(2)(9.27 \times 10^{-24}) = 1.85 \times 10^{-23}\,\text{J/T}.$$

Divide:

$$B^* \approx 0.05\,\text{T}.$$

$$\boxed{B \lesssim 50\,\text{mT for weak-field regime.}}$$

(c)

Form $\vec{F} = \vec{I} + \vec{J}$, $F \in \{|I - J|, \ldots, I + J\}$. Square the sum:

$$\vec{F}^2 = \vec{I}^2 + \vec{J}^2 + 2\vec{I} \cdot \vec{J}.$$

Rearrange for the dot product:

$$\vec{I} \cdot \vec{J} = \tfrac{1}{2}\big[\vec{F}^2 - \vec{I}^2 - \vec{J}^2\big].$$

Take eigenvalues in $|F, M_F\rangle$:

$$\vec{I} \cdot \vec{J} = \tfrac{1}{2}[F(F+1) - I(I+1) - J(J+1)]\hbar^2.$$

Absorb $\hbar^2$ into A:

$$\langle A\vec{I} \cdot \vec{J}\rangle = \tfrac{1}{2}AK, \qquad \boxed{K = F(F+1) - I(I+1) - J(J+1).}$$

Treat Zeeman as perturbation in $|F, M_F\rangle$. Project via Landé:

$$\langle \vec{J}\rangle = \frac{\vec{F} \cdot \vec{J}}{F(F+1)\hbar^2}\vec{F}, \qquad \langle \vec{I}\rangle = \frac{\vec{F} \cdot \vec{I}}{F(F+1)\hbar^2}\vec{F}.$$

Write $\vec{J} = \vec{F} - \vec{I}$ and square:

$$\vec{J}^2 = \vec{F}^2 + \vec{I}^2 - 2\vec{F} \cdot \vec{J}.$$

Solve for $\vec{F}\cdot\vec{J}$:

$$\vec{F}\cdot\vec{J} = \tfrac{1}{2}[F(F+1) + J(J+1) - I(I+1)]\hbar^2.$$

Write $\vec{I} = \vec{F} - \vec{J}$ and square:

$$\vec{I}^2 = \vec{F}^2 + \vec{J}^2 - 2\vec{F}\cdot\vec{I}.$$

Solve for $\vec{F}\cdot\vec{I}$:

$$\vec{F}\cdot\vec{I} = \tfrac{1}{2}[F(F+1) + I(I+1) - J(J+1)]\hbar^2.$$

Take $\vec{B} = B\hat{z}$ so $\langle F_z\rangle = M_F\hbar$:

$$\Delta E_{\mathrm{Z}} = g_J\mu_B B\langle J_z\rangle - g_I\mu_N B\langle I_z\rangle.$$

Insert projected components:

$$\Delta E_{\mathrm{Z}} = \left[g_J\mu_B\,\frac{\vec{F}\cdot\vec{J}}{F(F+1)\hbar^2} - g_I\mu_N\,\frac{\vec{F}\cdot\vec{I}}{F(F+1)\hbar^2}\right] BM_F\hbar.$$

Drop the nuclear piece ($\mu_N/\mu_B \sim m_e/m_p \ll 1$):

$$\Delta E_{\mathrm{Z}} \approx g_J\mu_B BM_F\hbar\,\frac{\vec{F}\cdot\vec{J}}{F(F+1)\hbar^2}.$$

Substitute $\vec{F}\cdot\vec{J}$:

$$\Delta E_{\mathrm{Z}} = g_J\,\frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}\,\mu_B BM_F.$$

Read off g_F:

$$\boxed{g_F = g_J\,\frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}, \quad \Delta E = \tfrac{1}{2}AK + g_F\mu_B BM_F.}$$

(d)

For $J = 3/2, I = 3$, allowed $F \in \{|I-J|, \ldots, I+J\}$:

$$F \in \{3/2,\ 5/2,\ 7/2,\ 9/2\}.$$

Compute $I(I+1)$ and $J(J+1)$:

$$I(I+1) = 12, \qquad J(J+1) = 15/4.$$

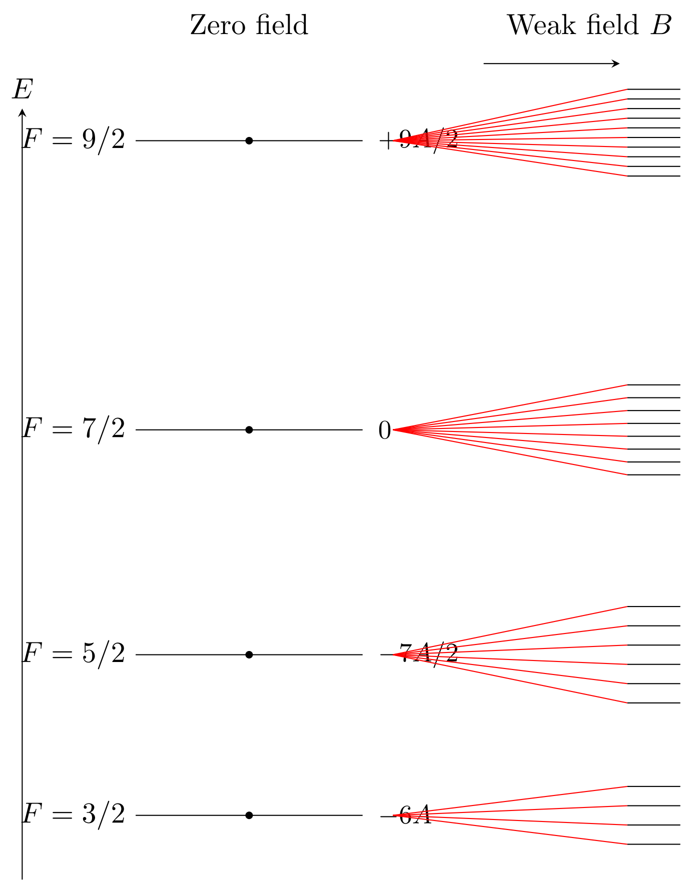
Form $K = F(F+1) - I(I+1) - J(J+1)$:

$$K = F(F+1) - 12 - 15/4.$$

Tabulate for each F:

F	$F(F+1)$	K	$\frac{1}{2}AK$
$3/2$	$15/4$	-12	$-6A$
$5/2$	$35/4$	-7	$-7A/2$
$7/2$	$63/4$	0	0
$9/2$	$99/4$	$+9$	$+9A/2$

Each F splits into $2F+1$ Zeeman sublevels: $4+6+8+10 = 28 = (2J+1)(2I+1)$. Sublevel spacing $g_F\mu_B B$ depends on F.



(e)

Strong field ($g_J\mu_B B \gg AI$): decouple, so J_z, I_z are good. For $1s\,^2S_{1/2}$ of H: $J = S = 1/2$, $I = 1/2$, $g_J \to g_s$, $g_I \to g_p$. Take $\vec{B} = B\hat{z}$ and evaluate each Hamiltonian term in $|M_J, M_I\rangle$:

$$\langle A\vec{I}\cdot\vec{J}\rangle = AM_JM_I, \quad \langle g_J\mu_B\vec{J}\cdot\vec{B}\rangle = g_s\mu_B BM_J, \quad -\langle g_I\mu_N\vec{I}\cdot\vec{B}\rangle = -g_p\mu_N BM_I.$$

Sum the three contributions:

$$\Delta E(M_J, M_I) = g_s\mu_B B\,M_J - g_p\mu_N B\,M_I + A\,M_JM_I.$$

Tabulate for the four states (M_J, M_I) with $M_J, M_I = \pm1/2$:

(M_J, M_I)	ΔE
$(+\frac{1}{2}, +\frac{1}{2})$	$\frac{1}{2}g_s\mu_B B - \frac{1}{2}g_p\mu_N B + \frac{1}{4}A$
$(+\frac{1}{2}, -\frac{1}{2})$	$\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B - \frac{1}{4}A$
$(-\frac{1}{2}, +\frac{1}{2})$	$-\frac{1}{2}g_s\mu_B B - \frac{1}{2}g_p\mu_N B - \frac{1}{4}A$
$(-\frac{1}{2}, -\frac{1}{2})$	$-\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B + \frac{1}{4}A$

Form numerator: add the two rows with $M_J = \pm1/2$, $M_I = -1/2$:

$$\Delta E_{+,-} + \Delta E_{-,-} = \left(\tfrac{1}{2} - \tfrac{1}{2}\right)g_s\mu_B B + \left(\tfrac{1}{2} + \tfrac{1}{2}\right)g_p\mu_N B + \left(-\tfrac{1}{4} + \tfrac{1}{4}\right)A.$$

Cancel coefficients:

$$\Delta E_{+,-} + \Delta E_{-,-} = 0 + g_p \mu_N B + 0.$$

Simplify:

$$\Delta E_{+,-} + \Delta E_{-,-} = g_p \mu_N B.$$

Form denominator: add the two rows with $M_J = +1/2$, $M_I = \pm 1/2$:

$$\Delta E_{+,-} + \Delta E_{+,+} = \left(\tfrac{1}{2} + \tfrac{1}{2}\right) g_s \mu_B B + \left(\tfrac{1}{2} - \tfrac{1}{2}\right) g_p \mu_N B + \left(-\tfrac{1}{4} + \tfrac{1}{4}\right) A.$$

Cancel coefficients:

$$\Delta E_{+,-} + \Delta E_{+,+} = g_s \mu_B B + 0 + 0.$$

Simplify:

$$\Delta E_{+,-} + \Delta E_{+,+} = g_s \mu_B B.$$

Divide numerator by denominator:

$$\boxed{\frac{\Delta E_{J_z=1/2, I_z=-1/2} + \Delta E_{J_z=-1/2, I_z=-1/2}}{\Delta E_{J_z=1/2, I_z=-1/2} + \Delta E_{J_z=1/2, I_z=1/2}} = \frac{g_p \mu_N}{g_s \mu_B}.}$$

Question 3 — Saturated gain and a spherical astrophysical maser

Problem. Two-level homogeneously broadened gain medium with densities $N_{1,2}$, degeneracies $g_{1,2}$, pump rates $R_{1,2}$, lifetimes $\tau_{1,2}$, lineshape $g(\omega - \omega_0)$, monochromatic laser $I(z,\omega) = I(z)\delta(\omega - \omega_L)$. (a) derive $\mathrm{d}I/\mathrm{d}z = \alpha_0 I/(1 + I/I_S)$ with α_0 and I_S in terms of Einstein coefficients; (b) for a uniformly inverted sphere, derive the radial growth equation; (c) given $I(R) = 0.1 I_S$, show $P(r) = 4\pi r^2 I(r)$ grows exponentially and find $I(r)$; (d) show $I(r)$ has a minimum, find its radius; (e) comment on $I(r = 0)$.

(a)

Net stimulated rate (gain − absorption) per unit volume per unit frequency:

$$W_{\text{stim}} = \left[N_2 B_{21} - N_1 B_{12}\right] u(\omega)\, g(\omega - \omega_0).$$

Apply the Einstein degeneracy relation $g_1 B_{12} = g_2 B_{21}$:

$$N_1 B_{12} = \frac{g_2}{g_1} N_1 B_{21}.$$

Substitute:

$$W_{\text{stim}} = \left[N_2 - \frac{g_2}{g_1} N_1\right] B_{21}\, u(\omega)\, g(\omega - \omega_0).$$

Energy gain per length: each net stimulated event adds $\hbar\omega_0$ to the beam:

$$\frac{\mathrm{d}I(z)}{\mathrm{d}z} = \hbar\omega_0 \int W_{\text{stim}}\, \mathrm{d}\omega.$$

For monochromatic $I(z,\omega) = I(z)\delta(\omega - \omega_L)$ the energy density is $u(\omega) = (I/c)\delta(\omega - \omega_L)$:

$$\frac{\mathrm{d}I(z)}{\mathrm{d}z} = \hbar\omega_0 \left[N_2 - \frac{g_2}{g_1} N_1\right] B_{21}\, g(\omega_L - \omega_0)\, \frac{I(z)}{c}.$$

Write rate equations under stimulated drive, lineshape $g \equiv g(\omega_L - \omega_0)$:

$$\dot{N}_2 = R_2 - \frac{N_2}{\tau_2} - \left[N_2 - \frac{g_2}{g_1}N_1\right]\frac{B_{21}I}{c}g,$$

$$\dot{N}_1 = R_1 - \frac{N_1}{\tau_1} + \left[N_2 - \frac{g_2}{g_1}N_1\right]\frac{B_{21}I}{c}g.$$

Define $\Delta N = N_2 - (g_2/g_1)N_1$ and impose steady state $\dot{N}_{1,2} = 0$:

$$N_2 = R_2\tau_2 - \tau_2\Delta N\,\frac{B_{21}Ig}{c}, \qquad N_1 = R_1\tau_1 + \tau_1\Delta N\,\frac{B_{21}Ig}{c}.$$

Form $N_2 - (g_2/g_1)N_1$:

$$\Delta N = R_2\tau_2 - \frac{g_2}{g_1}R_1\tau_1 - \left[\tau_2 + \frac{g_2}{g_1}\tau_1\right]\Delta N\,\frac{B_{21}Ig}{c}.$$

Collect ΔN on the left:

$$\Delta N\left[1 + \left(\tau_2 + \tfrac{g_2}{g_1}\tau_1\right)\frac{B_{21}Ig}{c}\right] = R_2\tau_2 - \frac{g_2}{g_1}R_1\tau_1.$$

Define $\Delta N_0 = R_2\tau_2 - (g_2/g_1)R_1\tau_1$ and $I_S = c/\{B_{21}g[\tau_2 + (g_2/g_1)\tau_1]\}$:

$$\Delta N = \frac{\Delta N_0}{1 + I/I_S}.$$

Read off the saturation intensity and small-signal gain:

$$\boxed{\alpha_0 = \frac{\hbar\omega_0}{c}\,\Delta N_0\,B_{21}\,g(\omega_L - \omega_0), \quad I_S = \frac{c}{B_{21}\,g(\omega_L - \omega_0)\big[\tau_2 + (g_2/g_1)\tau_1\big]}.}$$

Insert ΔN into dI/dz:

$$\boxed{\frac{\mathrm{d}I(z)}{\mathrm{d}z} = \frac{\alpha_0\,I(z)}{1 + I(z)/I_S}.}$$

(b)

Power crossing radius r: $P(r) = 4\pi r^2 I(r)$. Energy balance over shell $[r, r + \mathrm{d}r]$ with local gain $\mathrm{d}I/\mathrm{d}\ell = \alpha_0 I/(1 + I/I_S)$ along a ray:

$$\mathrm{d}P = \frac{\alpha_0\,I(r)}{1 + I(r)/I_S}\,4\pi r^2\,\mathrm{d}r.$$

Differentiate $P = 4\pi r^2 I$:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = 4\pi r^2\frac{\mathrm{d}I}{\mathrm{d}r} + 8\pi r I.$$

Equate the two expressions and divide by $4\pi r^2$:

$$\boxed{\frac{\mathrm{d}I}{\mathrm{d}r} + \frac{2I}{r} = \frac{\alpha_0\,I}{1 + I/I_S}.}$$

(c)

Assume the unsaturated regime $I \ll I_S$ holds throughout the sphere (consistent with $I(R) = 0.1 I_S$; the breakdown near $r = 0$ is addressed in (e)). Set $1 + I/I_S \approx 1$:

$$\frac{\mathrm{d}I}{\mathrm{d}r} + \frac{2I}{r} = \alpha_0 I.$$

Multiply through by $4\pi r^2$:

$$4\pi r^2 \frac{\mathrm{d}I}{\mathrm{d}r} + 8\pi r I = \alpha_0 \cdot 4\pi r^2 I.$$

Recognise the LHS as $\mathrm{d}P/\mathrm{d}r$:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = \alpha_0 P.$$

Integrate from r to R:

$$P(R) = P(r)\, e^{\alpha_0 (R-r)}.$$

Rearrange:

$$P(r) = P(R)\, e^{-\alpha_0 (R-r)}.$$

Use $P(R) = 4\pi R^2 I(R)$ and $I(r) = P(r)/(4\pi r^2)$:

$$\boxed{I(r) = I(R)\left(\frac{R}{r}\right)^2 e^{-\alpha_0 (R-r)} = 0.1\, I_S \left(\frac{R}{r}\right)^2 e^{-\alpha_0 (R-r)}.}$$

(d)

Rewrite $I(r) = 0.1 I_S R^2 e^{-\alpha_0 R} \cdot r^{-2} e^{\alpha_0 r}$ to isolate r-dependence:

$$\ln I = \text{const} - 2 \ln r + \alpha_0 r.$$

Take the logarithmic derivative:

$$\frac{1}{I}\frac{\mathrm{d}I}{\mathrm{d}r} = -\frac{2}{r} + \alpha_0.$$

Set $\mathrm{d}I/\mathrm{d}r = 0$ (equivalently the bracket vanishes):

$$-\frac{2}{r} + \alpha_0 = 0.$$

Solve for r:

$$\boxed{r_{\min} = \frac{2}{\alpha_0}.}$$

Differentiate $\ln I$ a second time:

$$\frac{\mathrm{d}^2 \ln I}{\mathrm{d}r^2} = \frac{2}{r^2} > 0.$$

Hence $\ln I$ (and I) has a minimum at $r = 2/\alpha_0$.

(e)

At $r = 0$ the formula $I(r) = 0.1 I_S (R/r)^2 e^{-\alpha_0 (R-r)}$ diverges as r^{-2}. Physically this is unphysical: (i) the geometric factor reflects an idealised point convergence of outward-going rays whose backward extrapolation crowds at the origin, (ii) once $I \to I_S$ the unsaturated approximation fails and saturation clamps the intensity, (iii) the spherical-wave $1/r^2$ assumption breaks below a wavelength. Take-away: in real masers the centre is heavily saturated and the actual $I(0)$ is bounded by $\sim I_S$, not infinite.

Question 4 — Semi-classical atom–field interaction

Problem. Two-level atom in classical field $V(t) = ex\mathcal{E}_0 \cos\omega t$, $\omega = \omega_0 + \delta\omega$, $|\delta\omega/\omega_0| \ll 1$. (a) derive the coupled amplitude equations; (b) RWA give $|c_2(t)|^2$; (c) derive $B_{12} = \pi\chi_{12}^2/(3\varepsilon_0\hbar^2)$; (d) compute B_{12} for hydrogen $1s \to 2p$.

(a)

Expand state in unperturbed eigenstates:

$$\Psi(\vec{r}, t) = c_1(t)\psi_1 e^{-iE_1 t/\hbar} + c_2(t)\psi_2 e^{-iE_2 t/\hbar}.$$

Insert into $i\hbar\partial_t\Psi = (H_0 + V)\Psi$ and use $H_0\psi_n = E_n\psi_n$:

$$i\hbar\big[\dot{c}_1\psi_1 e^{-iE_1 t/\hbar} + \dot{c}_2\psi_2 e^{-iE_2 t/\hbar}\big] = V(t)\big[c_1\psi_1 e^{-iE_1 t/\hbar} + c_2\psi_2 e^{-iE_2 t/\hbar}\big].$$

Project on ψ_1^* (orthonormality, $V_{11} = 0$ by parity):

$$i\hbar\dot{c}_1 = c_2 V_{12}(t) e^{-i\omega_0 t}, \qquad \omega_0 = (E_2 - E_1)/\hbar.$$

Project on ψ_2^* ($V_{22} = 0$ by parity):

$$i\hbar\dot{c}_2 = c_1 V_{21}(t) e^{+i\omega_0 t}.$$

Substitute $V = ex\mathcal{E}_0 \cos\omega t$ and use $\chi_{ab} = -e\int \psi_a^* x \psi_b \,\mathrm{d}\vec{r}$:

$$V_{ab}(t) = -\mathcal{E}_0 \cos(\omega t)\,\chi_{ab}.$$

Insert into $\dot{c}_1$:

$$i\hbar\dot{c}_1 = -\mathcal{E}_0\chi_{12}\cos(\omega t) c_2 e^{-i\omega_0 t}.$$

Expand cosine $\cos\omega t = \frac{1}{2}(e^{i\omega t} + e^{-i\omega t})$:

$$\dot{c}_1 = -\frac{\mathcal{E}_0\chi_{12}}{2i\hbar}\big[e^{i\omega t} + e^{-i\omega t}\big] c_2 e^{-i\omega_0 t}.$$

Use $-1/(2i) = i/2$ and combine exponentials:

$$\dot{c}_1 = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar}\big[e^{i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t}\big] c_2.$$

Likewise for $\dot{c}_2$, using $\chi_{21} = \chi_{12}^* = \chi_{12}$ (real ψ_n):

$$\boxed{\begin{aligned} \dot{c}_1(t) &= \frac{i\mathcal{E}_0\chi_{12}}{2\hbar}\big[e^{i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t}\big] c_2(t), \\ \dot{c}_2(t) &= \frac{i\mathcal{E}_0\chi_{12}}{2\hbar}\big[e^{-i(\omega-\omega_0)t} + e^{i(\omega+\omega_0)t}\big] c_1(t). \end{aligned}} \tag{1}$$

(b)

Initial $c_1(0) = 1$, $c_2(0) = 0$. Weak field: set $c_1 \approx 1$ in (1):

$$\dot{c}_2 = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar}\big[e^{-i(\omega-\omega_0)t} + e^{i(\omega+\omega_0)t}\big].$$

Drop the counter-rotating term (RWA, since $\omega + \omega_0 \sim 2\omega \gg |\omega - \omega_0|$):

$$\dot{c}_2 = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar} e^{-i(\omega-\omega_0)t}.$$

Integrate from 0 to t using $c_2(0) = 0$:

$$c_2(t) = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar}\int_0^t e^{-i(\omega-\omega_0)t'}\,\mathrm{d}t'.$$

Evaluate the integral:

$$c_2(t) = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar}\cdot\frac{e^{-i(\omega-\omega_0)t}-1}{-i(\omega-\omega_0)}.$$

Cancel $i/(-i) = -1$ and pull the sign in:

$$c_2(t) = -\frac{\mathcal{E}_0\chi_{12}}{2\hbar}\cdot\frac{e^{-i(\omega-\omega_0)t}-1}{\omega-\omega_0}.$$

Take square modulus:

$$|c_2(t)|^2 = \frac{\mathcal{E}_0^2\chi_{12}^2}{4\hbar^2}\,\frac{|e^{-i(\omega-\omega_0)t}-1|^2}{(\omega-\omega_0)^2}.$$

Use $|e^{i\theta}-1|^2 = 4\sin^2(\theta/2)$ with $\theta = -(\omega-\omega_0)t$:

$$|e^{-i(\omega-\omega_0)t}-1|^2 = 4\sin^2[(\omega-\omega_0)t/2].$$

Substitute:

$$\boxed{|c_2(t)|^2 = \frac{\mathcal{E}_0^2\chi_{12}^2}{\hbar^2}\,\frac{\sin^2[(\omega-\omega_0)t/2]}{(\omega-\omega_0)^2}.} \tag{2}$$

(c)

For a non-monochromatic field, the spectral content gives $\varepsilon_0\mathcal{E}_0^2(\omega)/2 = u(\omega)\mathrm{d}\omega$:

$$\mathcal{E}_0^2(\omega) = \frac{2u(\omega)\,\mathrm{d}\omega}{\varepsilon_0}.$$

Sum (2) incoherently over ω (each spectral component drives independently):

$$|c_2(t)|^2 = \int \frac{2\chi_{12}^2 u(\omega)}{\varepsilon_0\hbar^2}\,\frac{\sin^2[(\omega-\omega_0)t/2]}{(\omega-\omega_0)^2}\mathrm{d}\omega.$$

For large t the kernel becomes a delta (using $\int_{-\infty}^{\infty}\sin^2(xt/2)/x^2\,\mathrm{d}x = \pi t/2$):

$$\frac{\sin^2[(\omega-\omega_0)t/2]}{(\omega-\omega_0)^2} \to \frac{\pi t}{2}\,\delta(\omega-\omega_0).$$

Pull out slowly varying $u(\omega_0)$:

$$|c_2(t)|^2 = \frac{2\chi_{12}^2 u(\omega_0)}{\varepsilon_0\hbar^2}\cdot\frac{\pi t}{2}.$$

Cancel the 2:

$$|c_2(t)|^2 = \frac{\pi\chi_{12}^2}{\varepsilon_0\hbar^2}\,u(\omega_0)\,t.$$

Identify the absorption rate $|c_2|^2/t \equiv B_{12}u(\omega_0)$:

$$B_{12} = \frac{\pi\chi_{12}^2}{\varepsilon_0\hbar^2}.$$

Average over orientation of $\vec{E}$ relative to the atomic dipole, giving $\overline{\cos^2\theta} = 1/3$:

$$\boxed{B_{12} = \frac{\pi\chi_{12}^2}{3\varepsilon_0\hbar^2}.} \tag{3}$$

(d)

In (3), after orientation averaging, χ_{12}^2 represents the squared magnitude of the full dipole matrix element $|\vec{d}_{12}|^2 = e^2|\langle 1|\vec{r}|2\rangle|^2$. Sum over final m in the $2p$ manifold:

$$B_{12}^{1s\to 2p} = \frac{\pi e^2}{3\varepsilon_0\hbar^2}\sum_m \big|\langle 2p_m|\vec{r}|1s\rangle\big|^2.$$

By spherical symmetry the three Cartesian components contribute equally on summing over m:

$$\sum_m \big|\langle 2p_m|x|1s\rangle\big|^2 = \sum_m \big|\langle 2p_m|y|1s\rangle\big|^2 = \sum_m \big|\langle 2p_m|z|1s\rangle\big|^2.$$

Hence:

$$\sum_m \big|\langle 2p_m|\vec{r}|1s\rangle\big|^2 = 3\sum_m \big|\langle 2p_m|x|1s\rangle\big|^2.$$

Compute the x matrix element using $x = r\sin\theta\cos\phi$. Use the table conventions $Y_{1,-1} = (3/8\pi)^{1/2}\sin\theta e^{-i\phi}$ and $Y_{1,+1} = -(3/8\pi)^{1/2}\sin\theta e^{i\phi}$:

$$Y_{1,-1} - Y_{1,+1} = (3/8\pi)^{1/2}\sin\theta\big(e^{-i\phi} + e^{i\phi}\big).$$

Apply Euler's identity:

$$Y_{1,-1} - Y_{1,+1} = 2(3/8\pi)^{1/2}\sin\theta\cos\phi.$$

Solve for $\sin\theta\cos\phi$ and multiply by r:

$$x = r\sqrt{\tfrac{2\pi}{3}}\big(Y_{1,-1} - Y_{1,+1}\big).$$

Split the matrix element into radial and angular parts: $\langle 2p_m|x|1s\rangle = I_r A_m$ with

$$I_r = \int_0^\infty R_{10}(r)R_{21}(r)\,r^3\,\mathrm{d}r, \qquad A_m = \sqrt{\tfrac{2\pi}{3}}\int Y_{1,m}^*\big(Y_{1,-1} - Y_{1,+1}\big)Y_{0,0}\,\mathrm{d}\Omega.$$

Use $Y_{0,0} = 1/\sqrt{4\pi}$ and the orthonormality $\int Y_{1,m}^* Y_{1,m'}\,\mathrm{d}\Omega = \delta_{mm'}$:

$$\int Y_{1,m}^* Y_{1,-1}\,\mathrm{d}\Omega = \delta_{m,-1}, \qquad \int Y_{1,m}^* Y_{1,+1}\,\mathrm{d}\Omega = \delta_{m,+1}.$$

Combine the two integrals:

$$A_m = \frac{1}{\sqrt{4\pi}}\sqrt{\tfrac{2\pi}{3}}\big(\delta_{m,-1} - \delta_{m,+1}\big) = \frac{\delta_{m,-1} - \delta_{m,+1}}{\sqrt{6}}.$$

Read off values:

$$A_0 = 0, \quad A_{+1} = -\frac{1}{\sqrt{6}}, \quad A_{-1} = +\frac{1}{\sqrt{6}}.$$

Sum the squares:

$$\sum_m |A_m|^2 = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

Insert the radial functions from the table $R_{10} = 2a_0^{-3/2}e^{-r/a_0}$ and $R_{21} = (1/\sqrt{3})a_0^{-5/2}r\,e^{-r/2a_0}$:

$$R_{10}R_{21} = \frac{2}{\sqrt{3}}a_0^{-4}\,r\,e^{-3r/(2a_0)}.$$

Multiply by r^3:

$$R_{10}R_{21}r^3 = \frac{2}{\sqrt{3}}a_0^{-4}\,r^4\,e^{-3r/(2a_0)}.$$

Integrate:

$$I_r = \frac{2}{\sqrt{3}}a_0^{-4}\int_0^\infty r^4 e^{-3r/(2a_0)}\,\mathrm{d}r.$$

Use $\int_0^\infty r^n e^{-\alpha r}\mathrm{d}r = n!/\alpha^{n+1}$ with $\alpha = 3/(2a_0)$:

$$\int_0^\infty r^4 e^{-3r/(2a_0)}\mathrm{d}r = \frac{4!}{(3/2a_0)^5}.$$

Reduce the denominator:

$$\frac{4!}{(3/2a_0)^5} = \frac{24\cdot(2a_0)^5}{3^5} = \frac{24\cdot 32\,a_0^5}{243} = \frac{768\,a_0^5}{243}.$$

Substitute back:

$$I_r = \frac{2}{\sqrt{3}}a_0^{-4}\cdot\frac{768\,a_0^5}{243} = \frac{1536}{243\sqrt{3}}\,a_0.$$

Rationalise the surd:

$$I_r = \frac{1536\sqrt{3}}{729}\,a_0 = \frac{512\sqrt{3}}{243}\,a_0.$$

Square:

$$I_r^2 = \frac{512^2\cdot 3}{243^2}\,a_0^2 = 3\left(\frac{512}{243}\right)^2 a_0^2.$$

Form $\sum_m |\langle 2p_m|x|1s\rangle|^2 = I_r^2\sum_m |A_m|^2$:

$$\sum_m \left|\langle 2p_m|x|1s\rangle\right|^2 = 3\left(\frac{512}{243}\right)^2 a_0^2\cdot\frac{1}{3} = \left(\frac{512}{243}\right)^2 a_0^2.$$

Multiply by 3 for the vector sum:

$$\sum_m \left|\langle 2p_m|\vec{r}|1s\rangle\right|^2 = 3\left(\frac{512}{243}\right)^2 a_0^2.$$

Insert into the Einstein-B formula:

$$B_{12}^{1s\to 2p} = \frac{\pi e^2}{3\varepsilon_0\hbar^2}\cdot 3\left(\frac{2^9}{3^5}\right)^2 a_0^2.$$

Cancel the 3:

$$B_{12}^{1s\to 2p} = \frac{\pi\,e^2 a_0^2}{\varepsilon_0\hbar^2}\left(\frac{2^9}{3^5}\right)^2.$$

Plug in $e = 1.602\times10^{-19}\,\mathrm{C}$, $a_0 = 5.292\times10^{-11}\,\mathrm{m}$, $\varepsilon_0 = 8.854\times10^{-12}\,\mathrm{F/m}$, $\hbar = 1.055\times10^{-34}\,\mathrm{J\,s}$.

Square the elementary charge:

$$e^2 = 2.566\times10^{-38}\,\mathrm{C}^2.$$

Square the Bohr radius:

$$a_0^2 = 2.800\times10^{-21}\,\mathrm{m}^2.$$

Multiply:

$$e^2a_0^2 = 7.185\times10^{-59}\,\mathrm{C}^2\,\mathrm{m}^2.$$

Square $\hbar$:

$$\hbar^2 = 1.113 \times 10^{-68}\,\mathrm{J}^2\mathrm{s}^2.$$

Multiply by ε_0:

$$\varepsilon_0 \hbar^2 = 9.857 \times 10^{-80}.$$

Form the dimensional ratio:

$$\frac{e^2 a_0^2}{\varepsilon_0 \hbar^2} = \frac{7.185 \times 10^{-59}}{9.857 \times 10^{-80}} = 7.29 \times 10^{20}\ \mathrm{m}^3\,\mathrm{J}^{-1}\,\mathrm{s}^{-2}.$$

Evaluate the dimensionless factor:

$$\left(\frac{512}{243}\right)^2 = \frac{262144}{59049} \approx 4.440.$$

Combine:

$$B_{12}^{1s\to 2p} = \pi \cdot 4.440 \cdot 7.29 \times 10^{20}.$$

Multiply:

$$\boxed{B_{12}^{1s\to 2p} \approx 1.0 \times 10^{22}\,\mathrm{m}^3\,\mathrm{J}^{-1}\,\mathrm{s}^{-2}.}$$

Note. The $n = 2$ radial functions in the exam table are written without the standard normalisation factor of $1/(2\sqrt{2})$; using them verbatim makes $|R_{21}|^2$ a factor 8 too large. With correctly normalised wavefunctions one obtains the textbook value $B_{12}^{1s\to 2p} \approx 1.3\times 10^{21}\,\mathrm{m}^3\,\mathrm{J}^{-1}\,\mathrm{s}^{-2}$, consistent with $A_{2p\to 1s} = 6.27 \times 10^8\,\mathrm{s}^{-1}$ for Lyman-α.